Here $x_1 = 3,\ y_1 = 5,\ x_2 = 8,\ y_2 = -10$ and $m = 3,\ n = 2$

Therefore $P(x, y) = P\left(\frac{3(8)+2(3)}{3+2}, \frac{3(-10)+2(5)}{3+2}\right) = P\left(\frac{24+6}{5}, \frac{-30+10}{5}\right) = P(6, -4)$

Example 5.18 In what ratio does the point $P(-2, 4)$ divide the line segment joining the points $A(-3, 6)$ and $B(1, -2)$ internally?

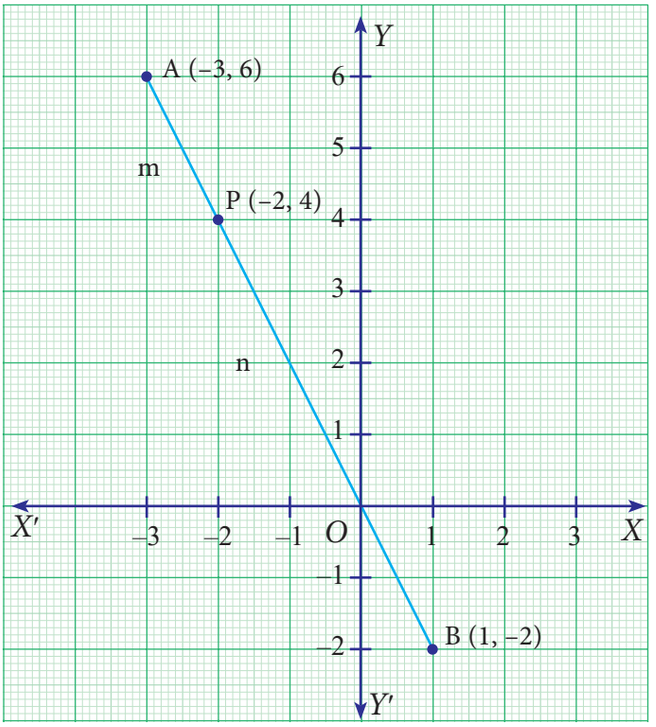


Fig. 5.36

Solution

Given points are $A(-3, 6)$ and $B(1, -2)$. $P(-2, 4)$ divide AB internally in the ratio $m : n$.

By section formula,

$$P(x, y) = P\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right)$$

$$= P(-2, 4) \quad .....(1)$$

Here $x_1 = -3,\ y_1 = 6,\ x_2 = 1,\ y_2 = -2$

$$(1) \Rightarrow \left(\frac{m(1)+n(-3)}{m+n}, \frac{m(-2)+n(6)}{m+n}\right) = P(-2, 4)$$

Equating x-coordinates, we get

$$\frac{m-3n}{m+n} = -2 \text{ or } m - 3n = -2m - 2n$$

$$3m = n$$

$$\frac{m}{n} = \frac{1}{3}$$

$$m : n = 1 : 3$$

Hence P divides AB internally in the ratio 1:3.

Note

We may arrive at the same result by also equating the y-coordinates.

Try it.

Example 5.19 What are the coordinates of B if point $P(-2,3)$ divides the line segment joining $A(-3,5)$ and B internally in the ratio 1:6?

Solution

Let $A(-3,5)$ and $B(x_2, y_2)$ be the given two points.

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